Venkata Lakshmi Parimala Pasupuleti

Senior Software Developer · LIHTC & Affordable Housing Domain · PHP/Laravel · Angular · .NET Framework · Microsoft SQL Server · Azure · Power Platform · 9yr Production

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PROFESSIONAL SUMMARY

Senior software developer with 9+ years of production experience and 7 years of direct LIHTC and affordable housing domain expertise - the precise combination CREA's technology team needs. At Entrata I owned the HUD income-limit import pipeline for nearly 7 years: annual federal data ingestion and normalization (few thousand source rows → 180,000 domain records) powering LIHTC eligibility, rent calculations, HAP, and tax-credit determinations for 100K+ housing units - with zero compliance violations across the full ownership period. I built and maintained this infrastructure in PHP, with Angular on the frontend, Microsoft SQL Server for early database work, and AWS/Azure cloud services throughout. I also bring .NET Framework experience from early-career web development, Power Apps from undergraduate work, and Azure cloud service integration from graduate research. Beyond the LIHTC domain, I bring full-stack product delivery across 120+ Agile sprints, RabbitMQ-based async processing, Docker microservices architecture, CI/CD automation, and data pipeline engineering at production scale. The system design and integration patterns CREA needs - PHP/Laravel backend, Angular frontend, MSSQL data layer, Azure services, Power Automate workflows - are not new territory. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages & Frameworks: PHP (Laravel), JavaScript, TypeScript, Angular, Python, SQL, PL/SQL, Bash, HTML/CSS, jQuery, AJAX, React.js, .NET Framework (VSTO, web development)

Data & Integration: Microsoft SQL Server (T-SQL), PostgreSQL, MySQL, Oracle, NoSQL, ETL/ELT Pipelines, Data Modeling, Data Normalization, REST APIs, API Design, System Integration, Cross-Platform Data Pipelines, RabbitMQ, Async Processing, Idempotency, Performance Optimization, Query Optimization, Indexing

Cloud, Automation & DevOps: Azure (App Service, Azure SQL, cloud services), AWS (S3, EC2, RDS, Glue), Google Cloud Storage (GCP), Microsoft Power Apps, Power Automate, Docker, CI/CD (GitHub Actions), Microservices, Observability, Structured Logging, Monitoring, Git, Linux, Agile/Scrum, Technical Mentoring, Code Reviews

EXPERIENCE

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built multi-source ML prediction pipelines on AWS and Azure cloud services across 190+ countries (R^2=0.977); applied Isolation Forest anomaly detection across 3 research pipelines; designed Tableau/ggplot2 dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and distributed systems.

Entrata - Multifamily Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **LIHTC & HUD compliance systems - 7-year ownership** - owned the annual HUD income-limit import pipeline: few thousand federal source rows normalized to ~180,000 domain records powering LIHTC eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units; every release required baseline validation, reconciliation, and rollback preparation (a wrong import affects thousands of residents' eligibility); zero compliance violations across full ownership period. Direct domain analog to CREA's LIHTC syndication operations.
- **Full-stack production delivery** - shipped 20+ modules (PHP, JavaScript, Angular, TypeScript, PostgreSQL/MySQL) across 120+ Agile sprints; RabbitMQ async distributed processing (retry logic, idempotency guarantees) for rent-schedule updates across thousands of tenant records; 200+ SQL query optimizations (30% performance gain); Docker microservices redesign (50% scalability improvement); GitHub Actions CI/CD with unit, integration, and regression test suites.
- **Data pipelines, integrations & cloud migration** - led AWS S3 document migration with Libsodium-encrypted ETL pipelines and secure coding practices (100% PHI compliance, zero security incidents); built scalable, maintainable Python/Bash automation eliminating 80% of manual data operations; structured logging and monitoring for pipeline oversight; led root cause analysis on production incidents, reducing triage time 40%; mentored 5+ engineers; 300+ code reviews.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee (.NET · PHP · Microsoft SQL Server)

- Built property management and affordable housing modules in PHP and .NET Framework; data pipeline for 1.6M HUD/Tax Credit records using Microsoft SQL Server; AWS S3 migration resolving 90% of legacy storage constraints; 56 state-specific compliance document templates.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Agentic Systems & Data Pipelines

- Sole technical owner of a 0→1 AI platform: designed data pipeline architecture, built RAG retrieval and compliance automation layer, implemented evaluation infrastructure and CI validation gate; delivered to production in weeks with measurable outcomes.

PROJECTS

Leasel: Compliance & Regulatory Intelligence Platform - LIHTC Domain

Compliance · Affordable Housing · Data Pipelines · REST APIs · Full-Stack · Production

Python · LangGraph · FastAPI · TF-IDF RAG · SQLite · REST APIs · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

AI-powered compliance investigation platform for residential lease and regulatory workflows - directly analogous to CREA's LIHTC compliance and investor/developer data workflows. Ingestion pipeline normalizing messy third-party documents (OCR fallback, section-aware chunking), regulatory document retrieval with 100% citation correctness, and structured compliance investigation workflows. Core design: deterministic guardrails ensure human oversight on all regulated decisions - LLM reasons and extracts, human or deterministic logic makes every final compliance call. 93.6% precision, 87.9% F1 across 1,240 test cases on residential lease compliance data.

AI on FHIR: Full-Stack Application on Azure & AWS - Regulated Domain

Full Stack · Azure · AWS Lambda · REST APIs · HIPAA · PostgreSQL · Docker

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · Azure · PostgreSQL · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack application on Azure and AWS Lambda cloud infrastructure for a privacy-regulated domain - NLP-powered query translation (EN/ES) into structured FHIR API calls; HIPAA-compliant RBAC with complete audit trail; Docker containerization; PostgreSQL data layer with full observability stack. Demonstrates full-stack delivery on Azure cloud services in a compliance-first, regulated environment - the same engineering discipline CREA's platform requires. ~60% reduction in manual retrieval time.

AgentProbe: Developer Productivity Platform - CI/CD & Automation

FastAPI · CI/CD · GitHub Actions · TypeScript · REST APIs · Automation · Open Source

Python · FastAPI · React.js · TypeScript · ChromaDB · GitHub Actions · REST APIs | github.com/venkatalakshmiparimala/AgentProbe

Production developer platform: 15-endpoint FastAPI backend, React.js + TypeScript frontend, CI-integrated regression gate (GitHub Actions) automating quality validation before deployment, and structured observability across all system operations. Demonstrates the API design, CI/CD automation, testing infrastructure, and developer tooling skills directly applicable to CREA's application modernization and continuous improvement initiatives.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017